

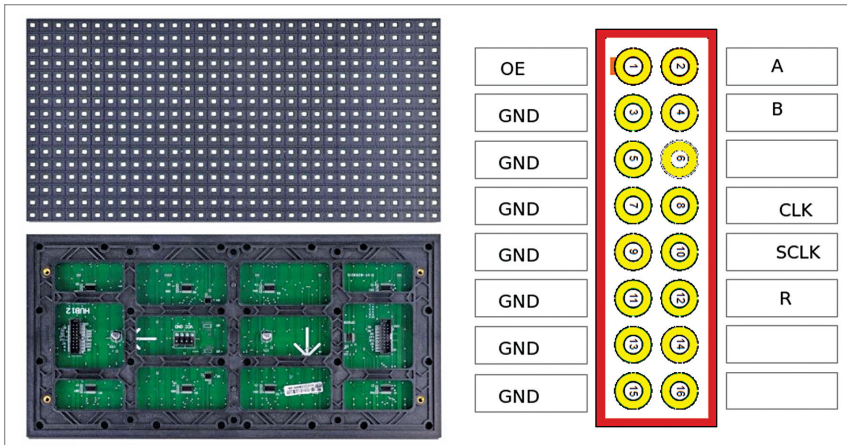


Fig. 3: P10 LED display and its pinouts for connection

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displays it on the LED matrix display.

The source code can be downloaded from <https://sourcecode.electronicsforu.com>. The code is uploaded by selecting the appropriate board and port in the IDE and connecting a laptop or desktop via USB. Fig. 4 shows a snippet of the source code.

Construction and testing

The source code is uploaded to the Arduino Mega before assembly. Once uploaded, the device is assembled based on the circuit shown in Fig. 2.

Switches SW1 through SW8 act as input switches for the eight players. These switches can be extended using twin wires for player convenience. Resistors R1 through R8 are used as pull-up resistors to keep the respective Arduino Mega pins high. Pressing switches SW1 through SW8 momentarily pulls the respective pins low.

The Arduino Mega is powered using a 9V power supply, and all resistors are connected to the 5V pin on the board. A 5V supply powers the P10 display.

When the push buttons (SW1 through SW8) are pressed, the system detects the first button pressed and displays the result. This design can be adapted to create a scoreboard for sports or an advertisement board. **EFY**



Palla Siva Kumari is lecturer in EEE department, Government Polytechnic, Kamalapuram, Andhra Pradesh.



Ramayanam Lavanya is pursuing Diploma in ECE Department, Government Polytechnic, Kamalapuram, Andhra Pradesh.

```
int a, b, c, d, e, f, g, x, i, j;
void setup() {
  Serial.begin(115200);
  pinMode(10, INPUT);
  pinMode(2, INPUT);
  pinMode(3, INPUT);
  pinMode(4, INPUT);
  pinMode(5, INPUT);
  pinMode(12, INPUT);
  pinMode(11, INPUT);
  pinMode(13, INPUT);
  Timer1.initialize(1000);
  Timer1.attachInterrupt(ScanDMD);
  dmd.clearScreen(true);
}
```

Fig. 4: Snippet of the source code

BILL OF MATERIALS

Components	Quantity
Arduino Mega (MOD1)	1
P10 LED display/dot matrix LED display (MOD2)	1
Push buttons (SW1-SW8)	8
1kΩ resistors (R1-R8)	8
5V, 2A power supply for P10 single colour display	1
Power supply (9V, 1A) for Mega	1
Jumper wires and wires	As required

Mega (MOD1), P10 LED display/dot matrix LED display (MOD2), push buttons (SW1-SW8), two power supplies—one for the P10 single-colour display and the other for the Arduino Mega—and a few other components.

The fastest finger first method involves candidates pressing a key placed in front of them. The first

person to press the key is granted the opportunity to answer the question.

In this setup, an Arduino Mega continuously monitors eight push buttons, each representing one of the eight candidates. The system identifies which button is pressed first. The button number of the candidate who presses the button first is displayed on an LED screen to indicate the winner. Pressing the reset button built into the Arduino Mega board returns the display to its initial state, ready for the next round.

The P10 display is connected to the Arduino Mega via SPI pins. Fig. 3 shows the P10 LED display and its pinouts for connection.

A P10 LED display module requires a suitable power supply to ensure optimal performance and longevity. P10 LED modules typically operate on 5V DC. It is essential to ensure the power supply delivers a stable and regulated 5V output. For small setups using one or two modules, a 5V, 2A SMPS is recommended.

Software

The source code is created in Arduino IDE. The fonts and other necessary files are included in the code. The input pins are configured to receive signals from the switches. In the main loop, the system checks which input is pressed first and